

A Project Report
on
Credit Card Fraud Detection
Random Forest Based Fraud Monitoring Dashboard

Submitted in partial fulfillment of the requirement of PROJECT
PROJECT-II (BCE5017)
of
Bachelor in Computer Engineering

Submitted to



Purbanchal University
Biratnagar, Nepal

Submitted By

Santosh Kumar Shah (751604)
Surya Chand Yadav (752074)
Salim Shrestha (751600)

Project Supervisor

Er. Sagar Dhakal

KANTIPUR CITY COLLEGE
Putalisadak, Kathmandu

A Project Report
on
Credit Card Fraud Detection

Submitted in partial fulfillment of the requirement of
PROJECT

PROJECT- II (BCE5017)
of
Bachelor in Computer Engineering

Submitted to
Purbanchal University
Biratnagar, Nepal

Submitted By
Santosh Kumar Shah (751604)
Surya Chand Yadav (752074)
Salim Shrestha (751600)

Project Supervisor
Er. Sagar Dhakal

KANTIPUR CITY COLLEGE
Putalisadak, Kathmandu

Declaration

We hereby declare that the project report entitled “*Credit Card Fraud Detection*”, submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering, is the result of our original work carried out under the supervision of **Er. Sagar Dhakal**.

We further declare that:

1. This work has not been submitted previously for the award of any degree, diploma, or similar title at any other institution or university. Proper acknowledgements have been given wherever the work of others has been referenced.
2. The project complies with the rules and regulations of Purbanchal University and Kantipur City College.

We understand that any violation of academic integrity may result in disciplinary action.

Santosh Kumar Shah
Surya Chand Yadav
Salim Shrestha

Acknowledgement

We would like to express our sincere gratitude to all those who supported and guided us in the successful completion of this project titled “*Credit Card Fraud Detection*”.

First and foremost, we extend our appreciation to our supervisor **Er. Sagar Dhakal** for his guidance, feedback, and support throughout the development and documentation of this project.

We also thank the faculty members and staff of the Department of Computer Engineering at Kantipur City College for providing the academic environment, resources, and suggestions needed to plan, implement, test, and document this work. We are grateful to our classmates, friends, and families for their cooperation, patience, and encouragement throughout the project period.

Santosh Kumar Shah
Surya Chand Yadav
Salim Shrestha

Abstract

This project presents a credit card fraud detection system named Fraud Shield. The system combines a trained Random Forest classifier with a Flask-based monitoring dashboard for real-time and batch transaction analysis. Unlike a standalone machine learning script, the application includes operational features that are important in fraud monitoring: authentication, transaction preprocessing, behavior profiling, risk scoring, alert generation, verification workflow, dashboard metrics, CSV upload, and downloadable reports.

The model uses 15 engineered features derived from transaction time, amount, merchant category, channel, country, and customer behavior. Raw transaction records are transformed by a shared feature engineering module, scaled using a fitted `StandardScaler`, and classified by a `RandomForestClassifier` saved as `fraud_model.pkl`. The application computes a fraud probability, converts it into a model decision using the configured threshold, combines it with behavior-based signals, and produces a risk score between 0 and 100. The system supports both single transaction checks through the dashboard/API and CSV batch detection through a manual upload workflow.

The training data stored in `creditcard_raw.csv` contains 59,978 transactions, including 922 fraud records, giving a fraud rate of approximately 1.54 percent. The current repository metadata reports a chronological 70/15/15 train-validation-test split and a recommended fraud probability threshold of approximately 0.449. On the test split, the model records 98.20 percent accuracy, 42.31 percent fraud precision, 47.83 percent fraud recall, 52.16 percent PR-AUC, and 94.96 percent ROC-AUC. This documentation is based on the implemented repository and related credit card fraud detection studies on imbalanced transaction classification and Random Forest based detection [1–3].

Keywords: Credit card fraud detection, Random Forest, machine learning, Flask dashboard, risk scoring, transaction monitoring, behavior analysis, fraud alerts.

Supervisor's Approval

This is to certify that the project entitled “*Credit Card Fraud Detection*”, undertaken and successfully demonstrated by **Santosh Kumar Shah, Surya Chand Yadav, and Salim Shrestha**, has been completed under my guidance.

This project is submitted as partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering under Purbanchal University. Throughout the duration of the project, the students have demonstrated dedication, technical understanding, and practical implementation skills in machine learning, web application development, testing, and documentation.

I hereby approve this project for certification by the concerned authority.

Er. Sagar Dhakal

Project Supervisor

Kantipur City College

Certificate from Department

This is to certify that, following the Supervisor's Approval and Examiners' Acceptance, the project entitled "*Credit Card Fraud Detection*", submitted by **Santosh Kumar Shah, Surya Chand Yadav**, and **Salim Shrestha**, has been officially approved as a partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering under Purbanchal University.

The department acknowledges the students' efforts and successful completion of the project.

Er. Subash Rajkarnikar

Head of Department,
Department of Computer Engineering,
Kantipur City College

Contents

Declaration	ii
Acknowledgement	iii
Abstract	iv
Supervisor’s Approval	v
Certificate from Department	vi
1 Introduction	1
1.1 Overview	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Major Features	2
1.5 Significance	3
1.6 Scope	3
1.7 Limitations	3
1.8 Organization of the Document	4
2 Literature Review	5
2.1 Theoretical Review	5
2.2 Empirical Review	5
2.3 Machine Learning Techniques for Fraud Detection	6
2.4 Research Gaps	6
2.5 Literature Summary	6
3 System Analysis	8
3.1 Existing System	8
3.2 Proposed System	8
3.3 Functional Requirements	8
3.4 Non-Functional Requirements	9

3.5	Feasibility Study	9
3.5.1	Technical Feasibility	9
3.5.2	Economic Feasibility	9
3.5.3	Operational Feasibility	10
3.5.4	Schedule Feasibility	10
3.6	Ethical and Security Considerations	10
4	Methodology	11
4.1	Software Development Life Cycle	11
4.2	Technology and Tools Used	12
4.3	Assignment of Roles and Responsibilities	12
4.4	Dataset	12
4.5	Data Preparation	13
4.6	Input Fields	13
4.7	Feature Engineering	13
4.8	Model Training	14
4.9	Prediction Pipeline	15
4.10	Risk Score Calculation	17
5	System Design	18
5.1	Architecture	18
5.2	Use Case Design	20
5.3	Data Flow Design	21
5.4	API Design	21
5.5	Persistence Design	22
5.6	User Interface Design	22
5.7	User Interface Screenshots	23
6	Implementation	26
6.1	Development Environment	26
6.2	Authentication	26
6.3	Transaction Validation	26
6.4	Feature Encoding	26
6.5	Behavior Analysis	27
6.6	Prediction and Risk	27
6.7	Alerts and Verification	27
6.8	CSV Batch Detection	27
6.9	Reporting	27
7	Testing and Results	28

7.1	Smoke Testing	28
7.2	Evaluation Metrics	29
7.3	Model Evaluation	29
7.4	Sample Input Results	30
7.5	Issues and Resolutions	31
7.6	Discussion	31
8	Conclusion and Future Work	32
8.1	Conclusion	32
8.2	Future Enhancements	32
A	Installation and User Guide	33
A.1	Environment Setup	33
A.2	Running the Application	33
A.3	CSV Batch Upload	34
A.4	Testing Command	34
	References	34

List of Figures

4.1	Prediction Workflow Flowchart	16
5.1	System Architecture	19
5.2	Use Case Diagram	20
5.3	Data Flow Diagram	23
5.4	Login Page	24
5.5	Transaction Prediction Form	24
5.6	Prediction Result View	25
5.7	CSV Batch Upload	25
5.8	Alert and Transaction Table	25

List of Tables

4.1	Technology and Tools Used	12
4.2	Roles and Responsibilities of Team Members	12
4.3	Model Features	14
4.4	Random Forest Training Configuration	15
5.1	Main Repository Components	21
5.2	Application Routes	21
7.1	Functional Test Cases	29
7.2	Confusion Matrix	30
7.3	Classification Report	30
7.4	Issues Encountered and Resolutions	31

Chapter 1

Introduction

1.1 Overview

Credit card fraud detection is a classification problem in which a system must identify whether a transaction is genuine or fraudulent. The difficulty of the task comes from class imbalance, changing fraud behavior, and the need for rapid decisions. The repository implements a practical fraud monitoring dashboard rather than only a notebook experiment. The main application is written in Flask and uses a trained Random Forest model to classify transactions.

The system is organized around the file `app.py`. It loads the model artifact `fraud_model.pkl` and the scaler artifact `scaler.pkl`, accepts transaction inputs, engineers the required features, predicts fraud probability, assigns a risk score, creates alerts, stores monitoring history, and exposes dashboard/reporting APIs. A shared module, `feature_engineering.py`, keeps training-time and inference-time feature definitions consistent.

1.2 Problem Statement

Manual inspection of card transactions is slow and unreliable when transaction volume is high. A fraud detection system must process transactions quickly, recognize risky behavior, reduce missed fraud cases, and still allow human review for high-risk or uncertain cases. The project addresses this problem by building an application that combines machine learning prediction with practical monitoring workflows.

The system must handle two types of input. First, an operator should be able to submit a single transaction through the dashboard and immediately see fraud probability, prediction label, risk score, behavior signals, and verification status. Second, an operator should be able to upload a CSV file containing multiple transactions and receive a processed summary

with errors, alerts, and updated dashboard metrics.

1.3 Objectives

- Develop a Random Forest based classifier for credit card fraud detection.
- Engineer meaningful transaction and behavior features instead of relying only on raw input fields.
- Build a Flask dashboard for authenticated fraud monitoring.
- Provide single transaction prediction and CSV batch fraud detection.
- Generate risk scores, alerts, transaction history, and downloadable CSV reports.
- Maintain consistency between model training and serving through shared feature definitions.
- Provide smoke tests that verify the main application routes and prediction workflow.

1.4 Major Features

- **Authentication:** Operators must sign in before accessing prediction and dashboard APIs.
- **Machine learning prediction:** A Random Forest classifier returns fraud probability and a binary label.
- **Behavior analysis:** Customer profiles track previous amounts, countries, devices, merchant categories, and recent transaction counts.
- **Risk scoring:** Model output and behavior signals are combined into a 0 to 100 risk score.
- **Real-time alerts:** Fraud predictions and high-risk transactions create alert records.
- **Verification workflow:** Operators can approve, block, review, or mark transactions as false positives.
- **CSV batch upload:** Multiple transactions can be processed from a structured CSV file.
- **Reports:** The system can export monitored transactions as `fraud_report.csv`.

1.5 Significance

The project is significant because credit card fraud can cause direct financial loss, customer inconvenience, and additional investigation workload for financial institutions. A system that can automatically score transactions helps analysts identify suspicious cases faster than manual review alone.

The implemented dashboard also demonstrates how a machine learning model can be converted into an operational workflow. Instead of stopping at model accuracy, the system provides fraud probability, risk level, behavior signals, alerts, verification status, CSV batch processing, and downloadable reports. These features make the project more practical for fraud monitoring because they support both automated screening and human decision-making.

For academic purposes, the project shows the complete path from data preparation and feature engineering to model training, deployment, testing, and documentation. It can also serve as a base for future work using real banking data, stronger security controls, database-backed persistence, and model drift monitoring.

1.6 Scope

The scope of this project includes the training and serving of a Random Forest model, a web dashboard, authenticated APIs, persistence through a JSON store, and test coverage through the included smoke-test script. The implementation is suitable for academic demonstration and for explaining how a fraud monitoring workflow can be built around a machine learning model.

1.7 Limitations

- The repository uses `instance/data_store.json` for runtime persistence instead of a production database.
- Authentication uses simple username/password checking and should be strengthened before production use.
- The dataset in the repository is synthetic and may not capture all real banking fraud patterns.
- Model drift, audit logging, encryption, rate limiting, and role-based access control are outside the current implementation.

- The model threshold is configurable through `FRAUD_THRESHOLD`; when it is not set, the application loads the recommended threshold from `model_metadata.json`.

1.8 Organization of the Document

Chapter 2 presents the literature and conceptual background. Chapter 3 describes the requirements and feasibility. Chapter 4 explains the methodology, feature engineering, model training, and prediction workflow. Chapter 5 presents the system design. Chapter 6 describes implementation details from the repository. Chapter 7 summarizes testing and results. Chapter 8 concludes the project and lists future enhancements. The appendix provides installation and user-guide instructions.

Chapter 2

Literature Review

2.1 Theoretical Review

Credit card fraud detection is commonly modeled as a supervised binary classification task. Each transaction is represented by features such as amount, time, merchant behavior, channel, and geographical indicators. The target class indicates whether the transaction is fraudulent. A key challenge is class imbalance: fraud transactions usually represent only a small fraction of the full dataset, so a model can obtain high accuracy while still missing important fraud cases.

Fraud detection systems usually combine transaction-level features and behavior-level features. Transaction-level features describe the current event, such as amount, hour, country, and channel. Behavior-level features compare the current event against the customer profile, such as average previous amount, recent transaction count, new country, and new merchant category. This combination is important because fraud is often visible as a deviation from previous customer behavior.

2.2 Empirical Review

Previous research on credit card fraud detection has applied many machine learning methods, including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting models, Autoencoders, and probabilistic ensemble models [4–6]. These studies commonly report that class imbalance is one of the main difficulties in fraud detection because normal transactions heavily outnumber fraudulent transactions [1, 2].

Studies also show that accuracy alone is not enough for evaluating fraud detection. A model can achieve high accuracy by predicting most transactions as normal, while still missing a large number of fraud cases. Therefore, precision, recall, F1-score, F2-score, PR-AUC,

ROC-AUC, and confusion matrix values are more meaningful for understanding fraud-class performance.

2.3 Machine Learning Techniques for Fraud Detection

Machine learning methods are useful because they can learn patterns from historical transaction data and generalize to new transactions. Logistic Regression is simple and interpretable, but may underfit nonlinear fraud patterns. Decision Trees are easy to explain, but can overfit. Support Vector Machines can work well on structured data, but may be harder to scale and tune. Neural networks can learn complex relationships, but require more data and careful training.

Random Forest is a suitable choice for this project because it is robust, handles nonlinear relationships, works well with tabular engineered features, and gives strong performance without requiring a very large deep-learning dataset [1, 2]. Random Forest is an ensemble learning method that builds many decision trees and combines their predictions. Each tree is trained on a bootstrapped sample of the data and considers a random subset of features at each split. This reduces overfitting compared with a single decision tree and improves generalization. In this project, `train_model.py` trains a `RandomForestClassifier` with 200 trees, class-weight balancing, maximum depth 12, and a fixed random seed for reproducibility.

2.4 Research Gaps

Many academic fraud detection projects focus mainly on training a model and reporting metrics. This is useful, but it does not fully represent the operational workflow needed by fraud analysts. In practice, analysts need a dashboard, authentication, transaction validation, risk scoring, alerts, verification actions, batch upload, and reports.

This project addresses that gap by connecting the machine learning model to a Flask-based monitoring dashboard. The repository extends model prediction with behavior profiling, risk calculation, alert generation, analyst verification, CSV batch detection, dashboard metrics, and report export. This makes the project more complete as a software system, not only as a model experiment.

2.5 Literature Summary

The reviewed work supports three design decisions in this project. First, fraud detection should be evaluated with fraud-sensitive metrics, not accuracy alone. Second, Random Forest is a practical baseline for tabular fraud features. Third, real-time and online payment

fraud systems require an operational layer around the model, including monitoring, alerts, and review workflows [3, 7]. This project applies those ideas through a working dashboard, behavior profiling, batch upload, and report generation.

Chapter 3

System Analysis

3.1 Existing System

Traditional fraud review relies on rule-based checks or manual investigation. These approaches can be useful, but they are limited when fraud behavior changes or transaction volume increases. Static rules may create many false positives or fail to identify new fraud patterns. Manual review also requires significant time and effort.

3.2 Proposed System

The proposed system is a web-based credit card fraud detector named Fraud Shield. It uses a Random Forest model for prediction and adds a monitoring layer around the model. Each transaction passes through validation, behavior analysis, feature engineering, scaling, model prediction, risk scoring, alert generation, profile update, and persistence. Operators interact with the system through a browser dashboard and API endpoints.

3.3 Functional Requirements

Requirement	Description
Login and logout	The system must authenticate operators before protected routes are accessed.
Single prediction	The system must accept transaction details and return fraud probability, label, risk score, and behavior signals.
CSV upload	The system must process CSV files containing multiple transactions.

Dashboard metrics	The system must show total transactions, fraud predictions, high-risk transactions, open alerts, verified transactions, and average risk.
Alert handling	The system must generate and acknowledge alerts for suspicious transactions.
Verification	The system must allow transactions to be approved, blocked, reviewed, or marked as false positives.
Report export	The system must provide a downloadable CSV report of monitored transactions.

3.4 Non-Functional Requirements

- **Usability:** The dashboard should be easy for an operator to use without command-line interaction.
- **Reliability:** Model and scaler artifacts must load successfully before prediction.
- **Maintainability:** Training and inference must share the same feature order and feature names.
- **Performance:** Predictions should be lightweight enough for real-time dashboard use.
- **Security:** Protected endpoints must reject unauthenticated requests.
- **Auditability:** Transaction records should include operator, verification status, behavior signals, and timestamps.

3.5 Feasibility Study

3.5.1 Technical Feasibility

The project is technically feasible because it uses well-supported Python libraries: Flask for the web application, Pandas and NumPy for data processing, Scikit-learn for machine learning, and Joblib for model persistence. The repository already contains trained artifacts and a smoke-test script, so the system can be run and tested locally.

3.5.2 Economic Feasibility

The tools used in this project are open source and do not require licensing fees. The system can run on a standard development machine, making it economically feasible for academic demonstration.

3.5.3 Operational Feasibility

The dashboard design supports operators by presenting prediction results, alerts, and verification actions in one place. CSV batch upload also makes the system practical for testing multiple transactions without writing API calls manually.

3.5.4 Schedule Feasibility

The project is schedule feasible because the work is divided into manageable phases: dataset preparation, feature engineering, model training, Flask application development, dashboard integration, testing, and report writing. The use of existing open-source libraries also reduces implementation time, allowing the main objectives to be completed within the Project-II semester timeline.

3.6 Ethical and Security Considerations

Fraud detection systems influence financial decisions, so their output should support human review rather than act as the only decision authority. In this project, high-risk transactions are placed in a verification workflow where an operator can approve, block, review, or mark the transaction as a false positive. This reduces the risk of treating every model prediction as final.

The current implementation is intended for academic demonstration and does not store real card numbers or sensitive customer records. A production deployment would require stronger controls, including hashed passwords, role-based access control, encrypted transport, audit logging, database access controls, and privacy review before using real banking data.

Chapter 4

Methodology

4.1 Software Development Life Cycle

The project follows an iterative development approach. Core functionality such as model training, prediction, and feature engineering was implemented first. Application features such as authentication, dashboard metrics, CSV upload, alert handling, verification, and reporting were then added and tested. This approach allowed each module to be verified before being connected to the complete fraud monitoring workflow.

4.2 Technology and Tools Used

Table 4.1: Technology and Tools Used

Technology / Tool	Purpose
Python	Main programming language for backend logic, data processing, and machine learning.
Flask	Web application framework used for dashboard routes, APIs, authentication, and request handling.
Scikit-learn	Machine learning library used for Random Forest training, scaling, metrics, and validation.
Pandas and NumPy	Data loading, preprocessing, feature computation, and numerical operations.
Joblib	Saving and loading trained model and scaler artifacts.
HTML, CSS, and JavaScript	Frontend dashboard interface and user interaction.
JSON	Lightweight runtime persistence for transactions, alerts, and behavior profiles.
LaTeX	Project report preparation and academic documentation.
Visual Studio Code	Development environment for editing, testing, and organizing source files.

4.3 Assignment of Roles and Responsibilities

Table 4.2: Roles and Responsibilities of Team Members

Team Member	Primary Responsibilities
Santosh Kumar Shah	Model training support, feature engineering review, prediction workflow testing, and documentation contribution.
Surya Chand Yadav	Flask dashboard support, CSV batch workflow, API testing, smoke-test verification, and result analysis.
Salim Shrestha	Report preparation, system design diagrams, frontend review, literature review, and final documentation formatting.

4.4 Dataset

The repository contains the dataset file `creditcard_raw.csv`. It has 59,978 transaction records. The target column is `is_fraud`. The dataset contains 922 fraud transactions, which is approximately 1.54 percent of the data. This imbalance is realistic for fraud

detection and is handled through model training choices such as class-weight balancing and validation-based threshold selection.

4.5 Data Preparation

The training script validates that required transaction columns are present before model training. Transaction date and time are converted into a combined timestamp, records are ordered chronologically, and customer-history features are calculated using only previous transactions. This prevents the model from using future behavior when computing features for earlier transactions.

Categorical fields such as merchant category, channel, and country are converted into fixed integer encodings. Unknown values are mapped to an `other` category. Amount values are converted to floating-point numbers and transformed with `log1p` to reduce the effect of very large transaction amounts.

4.6 Input Fields

The application requires the following transaction fields for prediction:

- `transaction_date`
- `transaction_time`
- `amount`
- `merchant_category`
- `country`
- `channel`

Optional metadata fields include `customer_id`, `card_last4`, `merchant`, and `device_id`. These fields support behavior analysis, dashboard display, and report generation.

4.7 Feature Engineering

The file `feature_engineering.py` defines the model feature list and the function `engineer_single_transaction()`. It converts raw transaction data and behavior signals into the exact 15 features expected by the trained model.

Table 4.3: Model Features

Feature	Meaning
hour	Hour extracted from transaction time.
day_of_week	Day index extracted from transaction date.
is_weekend	Indicates Saturday or Sunday.
is_night	Indicates transaction between 00:00 and 05:59.
amount	Transaction amount.
amount_log	Natural log transform of amount using $\log_{10}$.
merchant_category_enc	Encoded merchant category.
channel_enc	Encoded transaction channel.
country_enc	Encoded country.
txn_count_last_1h	Previous customer transactions in the last hour.
txn_count_last_24h	Previous customer transactions in the last 24 hours.
avg_amount_prev	Customer average amount before the current transaction.
amount_ratio	Current amount divided by previous average amount.
is_new_country	Indicates a country not previously seen for the customer.
is_new_category	Indicates a merchant category not previously seen for the customer.

4.8 Model Training

Training is implemented in `train_model.py`. The script loads `creditcard_raw.csv`, validates required columns, computes the engineered features, scales them with `StandardScaler`, splits the data using a chronological 70/15/15 train-validation-test split when all splits contain both classes, tunes a probability threshold on validation data, and trains a `RandomForestClassifier`.

Table 4.4: Random Forest Training Configuration

Parameter	Value
Model	RandomForestClassifier
Number of trees	200
Class weight	balanced
Maximum depth	12
Random state	42
Parallel jobs	-1
Split strategy	Chronological 70/15/15 train-validation-test split
Threshold selection	F2 score on validation data with minimum precision guard
Recommended threshold	0.448972
Cross-validation	5-fold TimeSeriesSplit on training data

4.9 Prediction Pipeline

The application prediction pipeline is implemented by `monitor_transaction()` in `app.py`. The workflow is:

1. Validate and sanitize the incoming transaction.
2. Load the customer behavior profile before updating it.
3. Analyze behavior signals such as new device, new country, high amount, and transaction velocity.
4. Engineer the 15 model features.
5. Scale features using `scaler.pkl`.
6. Predict fraud probability using `fraud_model.pkl`.
7. Convert probability into a binary prediction using the configured threshold.
8. Calculate risk score and risk level.
9. Create an alert when the transaction is model-flagged or high risk.
10. Update the customer behavior profile.
11. Save transaction, alert, and profile state to `instance/data_store.json`.

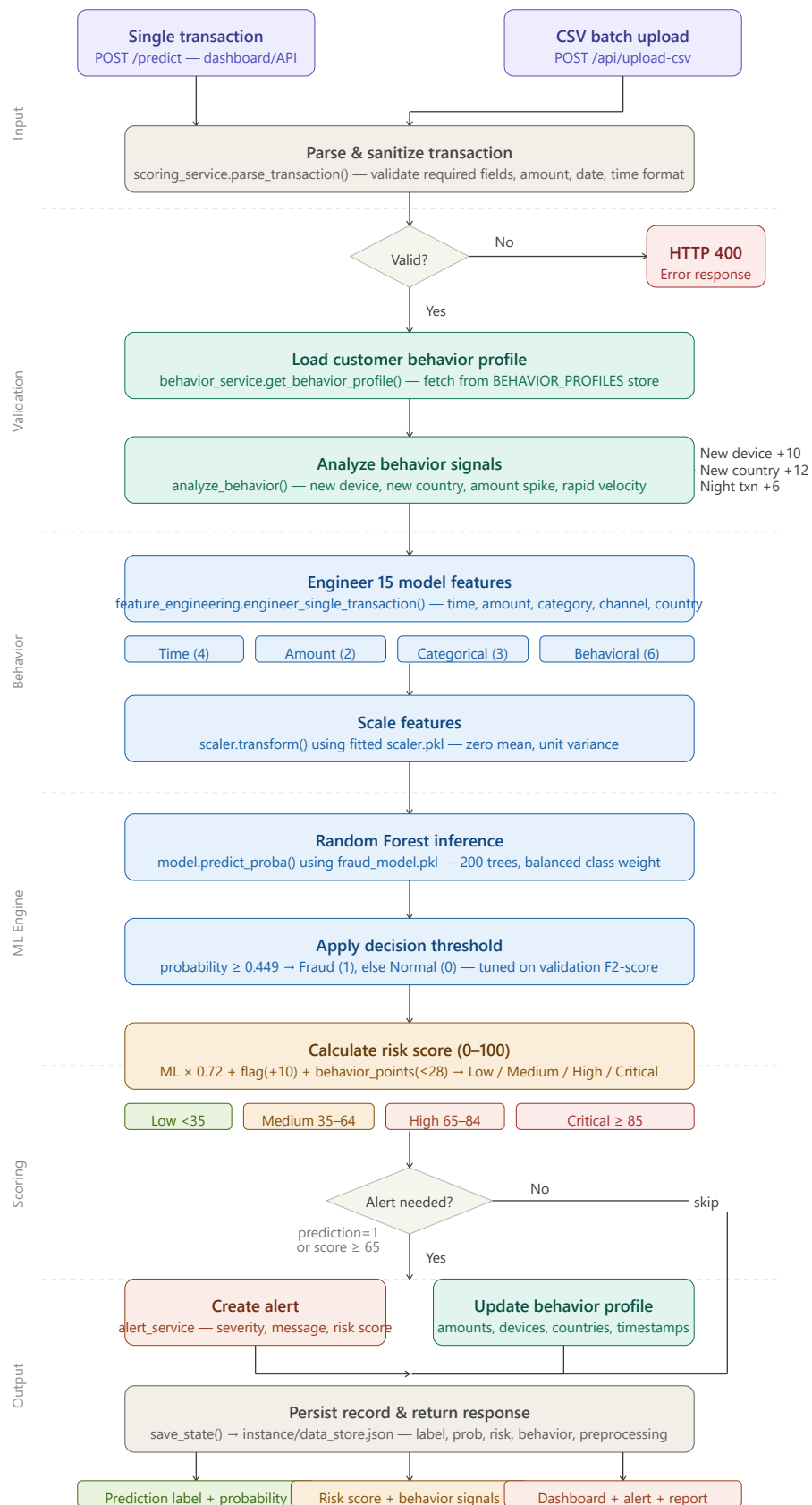


Figure 4.1: Prediction Workflow Flowchart

4.10 Risk Score Calculation

The risk score combines model probability, model decision, and behavior risk points. The implemented calculation is:

$$\text{risk score} = \min(100, \text{round}(72p + f + b))$$

where p is the model fraud probability, f is 10 when the model predicts fraud and 0 otherwise, and b is the behavior component capped at 28 points.

Risk levels are assigned as follows:

- Low: score below 35
- Medium: score from 35 to 64
- High: score from 65 to 84
- Critical: score 85 or above

Chapter 5

System Design

5.1 Architecture

The system follows a client-server architecture. The browser dashboard communicates with the Flask backend. The backend validates transactions, runs preprocessing, uses the machine learning artifacts, updates in-memory and JSON-based state, and returns JSON responses to the frontend.

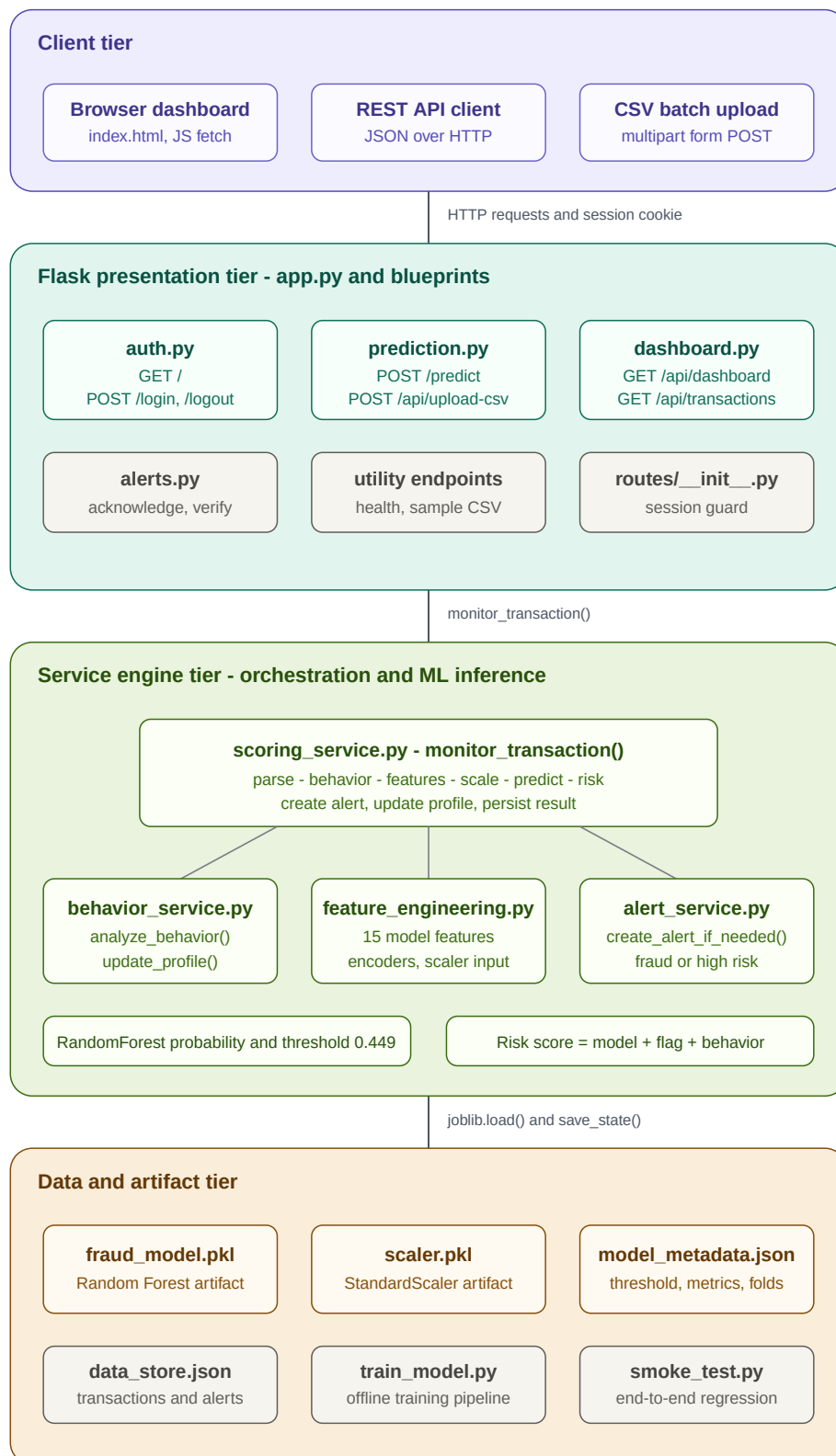


Figure 5.1: System Architecture

5.2 Use Case Design

The main actor of the system is the operator or administrator who reviews transactions and manages fraud alerts. The use case diagram in Figure 5.2 summarizes the primary dashboard functions available to this actor.

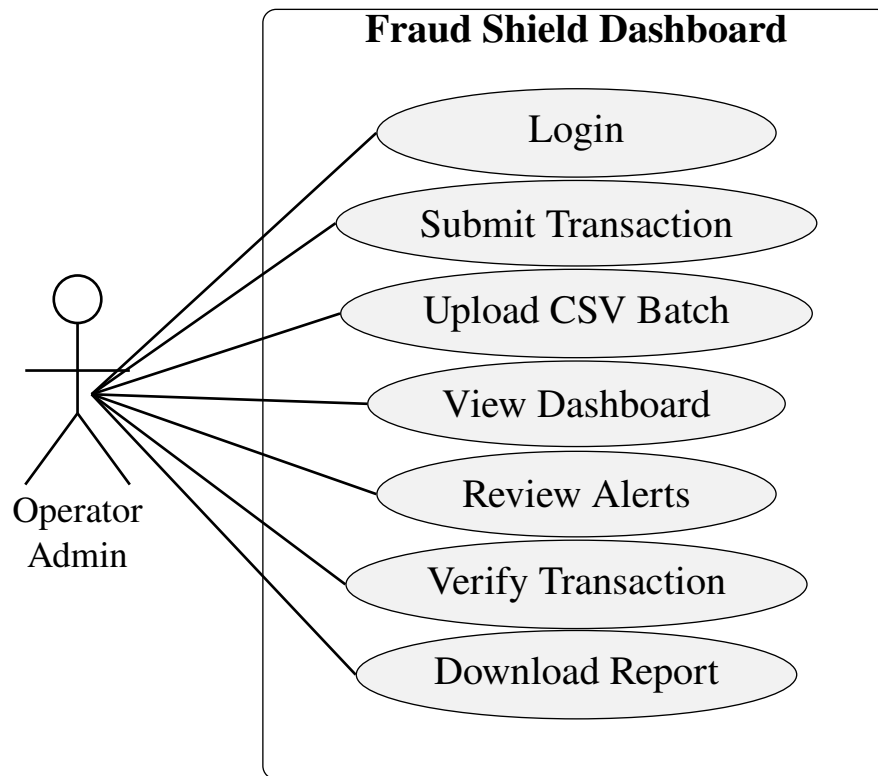


Figure 5.2: Use Case Diagram

Table 5.1: Main Repository Components

Component	Purpose
<code>app.py</code>	Flask server, routes, authentication, prediction pipeline, alerts, reports, and persistence.
<code>feature_engineering.py</code>	Shared feature definitions and single-transaction feature engineering.
<code>train_model.py</code>	Retraining script for model and scaler artifacts.
<code>fraud_model.pkl</code>	Trained Random Forest classifier.
<code>scaler.pkl</code>	Fitted StandardScaler for the 15 model features.
<code>templates/index.html</code>	Dashboard UI for login, prediction, CSV upload, alerts, and verification.
<code>smoke_test.py</code>	End-to-end checks for the Flask application.
<code>sample_upload.csv</code>	Example CSV file for batch detection.
<code>sample_transactions.json</code>	Example normal and fraud transaction payloads.
<code>instance/data_store.json</code>	Runtime persistence for transactions, alerts, and behavior profiles.

5.3 Data Flow Design

The data flow diagram shows how transaction data moves through the dashboard, backend, machine learning pipeline, alert workflow, and runtime storage. It complements the prediction workflow flowchart by showing the main data stores and outputs used by the complete monitoring system.

5.4 API Design

Table 5.2: Application Routes

Method	Endpoint	Purpose
GET	/	Render login page or dashboard.
POST	/login	Authenticate operator.
POST	/logout	Clear session.
GET	/health	Return application status, threshold, model files, and feature list.
POST	/predict	Process one transaction and return fraud/risk result.
GET	/api/dashboard	Return dashboard metrics, recent transactions, alerts, and profiles.

Method	Endpoint	Purpose
GET	/api/transactions	Return monitored transactions.
GET	/api/alerts	Return active alerts.
POST	/api/alerts/<alert_id>/acknowledge	Mark an alert as acknowledged.
POST	/api/transactions/<transaction_id>/verify	Update transaction verification status.
GET	/api/sample-csv	Download sample batch upload CSV.
POST	/api/upload-csv	Process uploaded CSV transactions.
GET	/api/report.csv	Download monitored transaction report.

5.5 Persistence Design

The application stores runtime state in memory and writes it to `instance/data_store.json`. If an older root-level `data_store.json` file exists, the application can read it as a legacy state file, but new writes use the `instance` directory. The state includes monitored transactions, alerts, and behavior profiles. Profiles contain transaction counts, amount statistics, recent timestamps, known devices, known countries, and known merchant categories. This design is simple and adequate for a demonstration project, but a production system should use a database.

5.6 User Interface Design

The dashboard in `templates/index.html` contains the following major areas:

- Login panel for unauthenticated users.
- Transaction monitoring form for single prediction.
- Prediction result cards for label, fraud probability, risk score, and verification status.
- CSV upload panel for batch fraud detection.
- Behavior signals and preprocessing summary.
- Alert queue and dashboard metrics.
- Recent transaction table with verification actions.

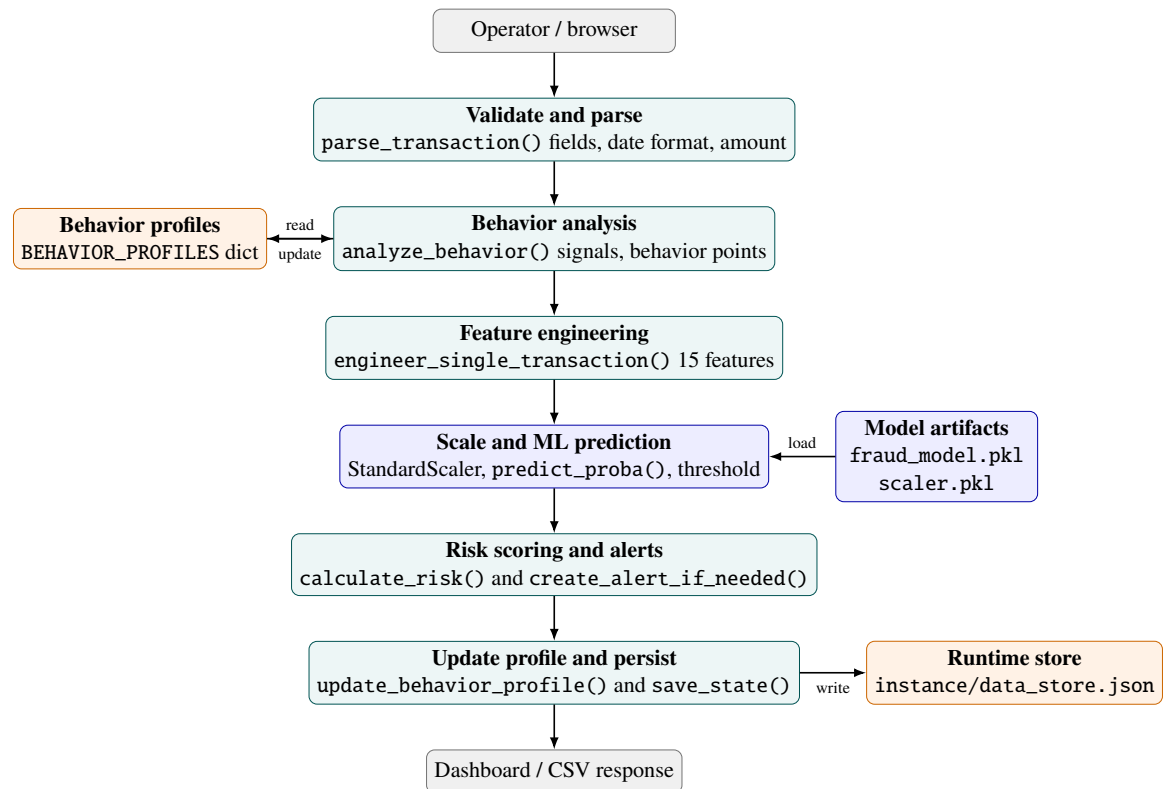
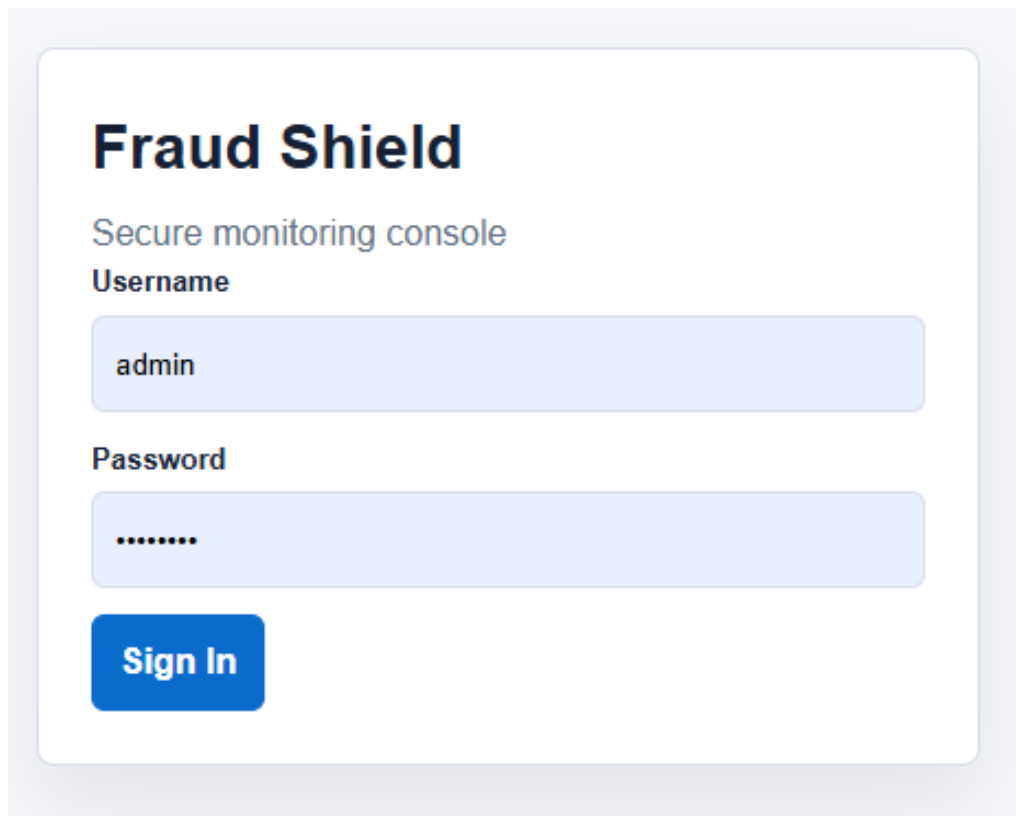


Figure 5.3: Data Flow Diagram

5.7 User Interface Screenshots

The following screenshots show the main operator-facing screens of the implemented dashboard.



The login page features a white card with rounded corners on a light gray background. At the top, the title 'Fraud Shield' is in a large, bold, dark blue font. Below it, the subtitle 'Secure monitoring console' is in a smaller, gray font. The 'Username' label is in bold, followed by a light blue input field containing the text 'admin'. The 'Password' label is also in bold, followed by a light blue input field with masked characters '.....'. A blue 'Sign In' button is positioned below the password field.

Fraud Shield

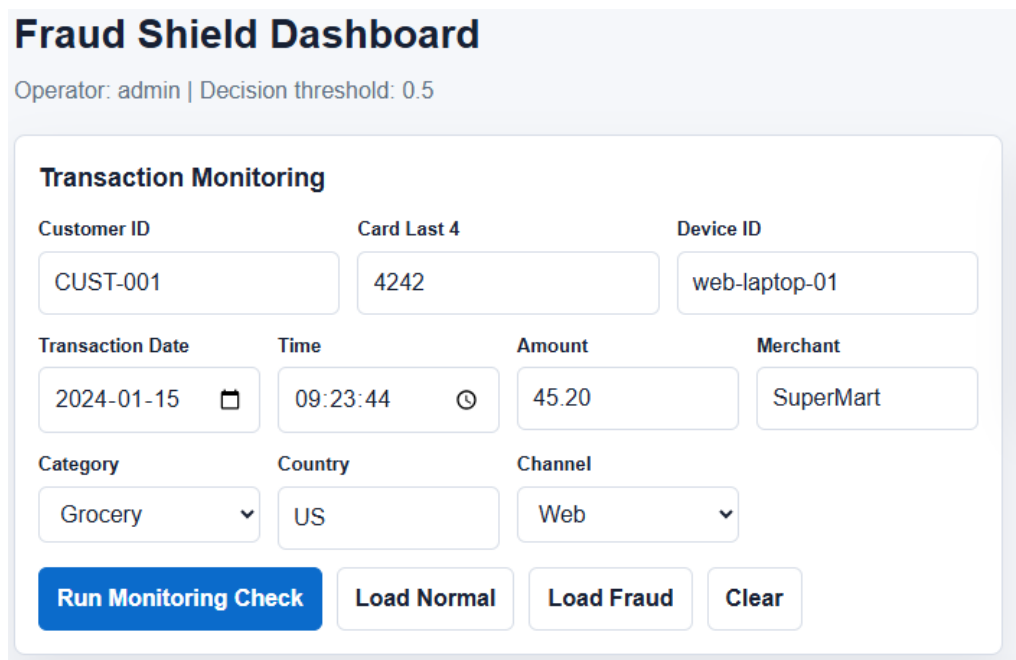
Secure monitoring console

Username

Password

Sign In

Figure 5.4: Login Page



The dashboard has a light gray header with the title 'Fraud Shield Dashboard' and a status bar showing 'Operator: admin | Decision threshold: 0.5'. The main section is a white card titled 'Transaction Monitoring'. It contains several input fields: 'Customer ID' (CUST-001), 'Card Last 4' (4242), and 'Device ID' (web-laptop-01). Below these are 'Transaction Date' (2024-01-15 with a calendar icon), 'Time' (09:23:44 with a clock icon), 'Amount' (45.20), and 'Merchant' (SuperMart). Further down are 'Category' (Grocery with a dropdown arrow), 'Country' (US), and 'Channel' (Web with a dropdown arrow). At the bottom are four buttons: 'Run Monitoring Check' (blue), 'Load Normal', 'Load Fraud', and 'Clear'.

Fraud Shield Dashboard

Operator: admin | Decision threshold: 0.5

Transaction Monitoring

Customer ID	Card Last 4	Device ID
CUST-001	4242	web-laptop-01

Transaction Date	Time	Amount	Merchant
2024-01-15 	09:23:44 	45.20	SuperMart

Category	Country	Channel
Grocery 	US	Web 

Run Monitoring Check

Figure 5.5: Transaction Prediction Form

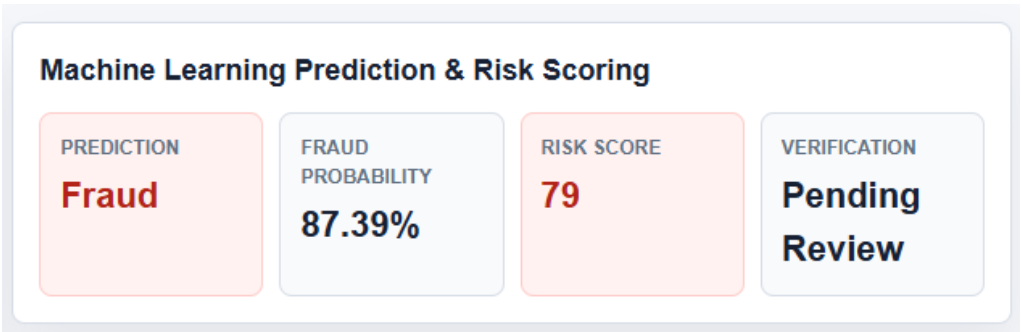


Figure 5.6: Prediction Result View



Figure 5.7: CSV Batch Upload

User Verification Queue								
TRANSACTION	CUSTOMER	AMOUNT	ML	RISK	STATUS	ACTION		
22ae2ab8 2026-05-20T17:58:40+00:00	CUST-001 SuperMart	45.20	Normal 0.00%	0 Low	Auto Cleared	Approve	Block	False Positive
3c214290 2026-05-20T17:58:37+00:00	CUST-009 Unknown ATM	500.00	Fraud 87.39%	79 High	Pending Review	Approve	Block	False Positive
47d1d073 2026-05-20T17:56:51+00:00	CUST-009 Unknown ATM	500.00	Fraud 87.39%	79 High	Pending Review	Approve	Block	False Positive
08c4390e 2026-05-20T16:57:39+00:00	CUST-001 SuperMart	45.20	Normal 0.00%	0 Low	Auto Cleared	Approve	Block	False Positive
d536b7b7 2026-05-20T16:57:37+00:00	CUST-009 Unknown ATM	500.00	Fraud 88.39%	80 High	Pending Review	Approve	Block	False Positive
69afb8f8 2026-05-20T16:15:19+00:00	CUST-009 Unknown ATM	500.00	Fraud 89.89%	81 High	Pending Review	Approve	Block	False Positive
e76bff0b 2026-05-20T16:13:25+00:00	CUST-009 Unknown ATM	500.00	Fraud 91.37%	82 High	Pending Review	Approve	Block	False Positive

Figure 5.8: Alert and Transaction Table

Chapter 6

Implementation

6.1 Development Environment

The project is implemented in Python. The required dependencies are listed in `requirements.txt`: Flask, Pandas, NumPy, Scikit-learn, and Joblib. The implementation is organized into separate files for the Flask application, feature engineering, model training, synthetic data generation, smoke testing, templates, and saved model artifacts.

6.2 Authentication

The application uses session-based authentication. The username, password, and secret key can be configured through environment variables. Protected endpoints use the `require_auth` decorator, which returns HTTP 401 for unauthenticated requests.

6.3 Transaction Validation

The function `parse_transaction()` validates required fields and ensures that `amount` is a non-negative finite number. It also fills optional metadata with safe defaults when values are missing. This prevents the model pipeline from receiving incomplete or malformed transaction objects.

6.4 Feature Encoding

The feature engineering module contains fixed category lists for merchant categories, channels, and countries. Unknown values are mapped to the final `other` category. The order of these lists is important because the encoded integer values are part of the model input.

6.5 Behavior Analysis

The function `analyze_behavior()` computes behavior points and signal descriptions. It checks whether the customer is new, whether the amount is much higher than previous average, whether the device, country, or merchant category is new, whether recent transaction count is high, whether the amount is large, and whether the transaction occurs at night.

6.6 Prediction and Risk

The `score_ml()` function calls `predict_proba()` on the Random Forest model and uses the configured threshold to decide whether the transaction is fraud. The `calculate_risk()` function combines model probability and behavior points to produce the final operational risk score. This means an unusual but model-benign transaction can still become high risk if behavior indicators are strong.

6.7 Alerts and Verification

The system creates an alert when the transaction is predicted as fraud or when the risk score is at least 65. High-risk transactions are placed in `PendingReview`. Operators can update verification status through the dashboard as `Approved`, `Blocked`, `Reviewed`, or `False Positive`. When a transaction is verified, related alerts are acknowledged.

6.8 CSV Batch Detection

The CSV upload endpoint reads an uploaded file with Pandas, enforces a maximum of 500 rows, checks required columns, and processes each row through the same monitoring pipeline used by single prediction. Each successful row returns transaction ID, customer ID, amount, label, fraud probability, risk score, risk level, and verification status.

6.9 Reporting

The report endpoint creates a CSV file with transaction ID, timestamp, customer ID, merchant, amount, fraud probability, model prediction, risk score, risk level, verification status, and behavior signals. This makes the dashboard output easy to preserve or analyze in spreadsheet software.

Chapter 7

Testing and Results

7.1 Smoke Testing

The repository includes `smoke_test.py`, which uses the Flask test client for end-to-end checks. The script verifies:

- `/health` returns status 200 and a valid feature count.
- Login succeeds with correct credentials and fails with wrong credentials.
- A normal sample transaction returns the expected Normal label.
- A fraud sample transaction returns the expected Fraud label.
- Fraud probability is within the range 0 to 1.
- Risk score is within the range 0 to 100.
- Behavior and preprocessing fields are present.
- Dashboard, sample CSV, CSV upload, report download, logout, and authentication checks work.

Table 7.1: Functional Test Cases

Test Case	Input / Action	Expected Result	Actual Result	Status
Feature encoding	Encode US and lower-case ru.	Correct country indexes are returned.	Indexes matched expected values.	Pass
Health check	Request /health.	HTTP 200 with status ok and feature count.	Endpoint returned valid health data.	Pass
Valid login	Submit correct username and password.	User is authenticated successfully.	Login returned status ok.	Pass
Invalid login	Submit wrong password.	Request is rejected with HTTP 401.	Login failed with HTTP 401.	Pass
Normal prediction	Submit normal sample transaction.	Label is Normal with valid score fields.	Normal label and valid scores returned.	Pass
Fraud prediction	Submit fraud sample transaction.	Label is Fraud with valid score fields.	Fraud label and valid scores returned.	Pass
Validation error	Submit invalid transaction date.	Request is rejected with HTTP 400.	Invalid date returned HTTP 400.	Pass
CSV upload	Upload sample_upload.csv.	Rows are processed with zero sample failures.	Records returned and failed count was zero.	Pass
Dashboard	Request authenticated dashboard API.	Metrics, transactions, and alerts are returned.	Dashboard payload was returned.	Pass
Report download	Request /api/report.csv.	Non-empty CSV response is downloaded.	CSV response was generated.	Pass
Logout protection	Logout and request dashboard again.	Protected route rejects unauthenticated user.	Dashboard returned HTTP 401.	Pass

7.2 Evaluation Metrics

The evaluation uses metrics that are suitable for imbalanced fraud detection:

- **Precision:** the proportion of predicted fraud transactions that are actually fraud.
- **Recall:** the proportion of actual fraud transactions detected by the model.
- **F1-score:** the harmonic mean of precision and recall.
- **F2-score:** a recall-weighted score that gives more importance to detecting fraud cases.
- **PR-AUC:** area under the precision-recall curve, useful when fraud cases are rare.
- **ROC-AUC:** area under the receiver operating characteristic curve.

7.3 Model Evaluation

The model performance reported in `model_metadata.json` is based on the current chronological test split. The dataset contains 59,978 synthetic transactions with 922 fraud transactions. The reported confusion matrix is shown below.

Table 7.2: Confusion Matrix

	Predicted Normal	Predicted Fraud
Actual Normal	8769	90
Actual Fraud	72	66

Table 7.3: Classification Report

Class	Precision	Recall	F1-score	Support
Normal	0.992	0.990	0.991	8859
Fraud	0.423	0.478	0.449	138
Accuracy		0.982		8997
Macro average	0.707	0.734	0.720	8997
Weighted average	0.983	0.982	0.983	8997

The repository metadata reports 5-fold time-series cross-validation on training data with a mean fraud recall of 0.3631 ± 0.1065 and a mean PR-AUC of 0.4938 ± 0.0407 . These results show that the model performs well for normal-class separation, but fraud-class recall remains an important improvement area because the dataset is highly imbalanced.

7.4 Sample Input Results

The repository includes two sample JSON payloads. The normal sample uses a grocery transaction of 45.20 in the United States. The fraud sample uses a cash transaction of 500.00 from Russia at night on an unknown device. The smoke test expects the first sample to be classified as Normal and the second as Fraud.

7.5 Issues and Resolutions

Table 7.4: Issues Encountered and Resolutions

Issue	Resolution
Class imbalance in fraud data	Used class-weight balancing in Random Forest training and selected a probability threshold using validation metrics.
Training and inference feature mismatch risk	Centralized feature names and single-transaction engineering in <code>feature_engineering.py</code> .
Unreliable first-transaction behavior features	Used neutral behavior values for new customer profiles so the current transaction does not distort its own score.
CSV input errors	Added required-column validation, row limits, date/time normalization, and per-row error reporting.
False positive handling	Added analyst verification statuses such as Approved, Blocked, Reviewed, and False Positive.
Runtime persistence limitation	Used JSON persistence for demonstration and listed relational database migration as future work.
Operational visibility	Added dashboard metrics, alerts, transaction history, and downloadable reports.

7.6 Discussion

The combination of model probability and behavior analysis is important. The Random Forest provides the statistical prediction, while behavior scoring provides operational context. This design supports practical fraud review by exposing both the model output and the reasons a transaction may be risky.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The project successfully implements a credit card fraud detection system using a Random Forest model and a Flask dashboard. The repository demonstrates an end-to-end workflow: data preparation, feature engineering, model training, prediction serving, risk scoring, alerting, verification, CSV batch detection, and reporting. By using shared feature definitions, the project reduces the risk of mismatch between model training and application inference.

The system is suitable for academic demonstration because it is runnable, testable, and connected to an operator-facing interface. The model evaluation shows strong performance on the available dataset, and the dashboard provides the workflow needed to inspect, verify, and report suspicious transactions.

8.2 Future Enhancements

- Replace JSON-file persistence with PostgreSQL or MySQL for reliable storage.
- Add hashed passwords, role-based access control, and stronger session security.
- Add audit logs for all alert and verification actions.
- Add model versioning and store which model scored each transaction.
- Monitor model drift and retrain with real-world transaction data.
- Add explainability features such as feature importance or per-transaction explanations.
- Add API rate limiting and HTTPS deployment configuration.
- Expand testing with unit tests for feature engineering and risk scoring.

Appendix A

Installation and User Guide

A.1 Environment Setup

The project requires Python and the packages listed in `requirements.txt`. From the project root directory, the environment can be prepared with:

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

A.2 Running the Application

The dashboard can be started with:

```
python app.py
```

After the Flask server starts, the dashboard is available at:

```
http://127.0.0.1:5000/
```

The default local login credentials are:

Username: admin

Password: admin123

For a real deployment, these credentials should be replaced using environment variables:

```
$env:APP_USERNAME = "analyst"
$env:APP_PASSWORD = "strong-password"
$env:SECRET_KEY = "replace-with-a-random-secret"
```

A.3 CSV Batch Upload

The CSV upload workflow expects transaction records with fields such as customer ID, card number suffix, merchant, merchant category, country, device ID, channel, transaction date, transaction time, and amount. A working example is provided in `sample_upload.csv`, and a downloadable sample is also exposed by the route `/api/sample-csv`.

A.4 Testing Command

The smoke test can be run from the project root with:

```
python smoke_test.py
```

References

- [1] Manoj Kumar Reddy Mallidi and Yeshwanth Zagabathuni. Analysis of credit card fraud detection using machine learning models on balanced and imbalanced datasets. *International Journal of Emerging Trends in Engineering Research*, 9(7):846–852, 2021.
- [2] Mayowa Timothy Adesina and Luke Howe. Credit card fraud detection. *International Journal of Science and Research Archive*, 12(2):2072–2080, 2024.
- [3] Faisal Tadvī, Sejal Shinde, Devendra Patil, and Sejal Dmello. Real time credit card fraud detection. *International Research Journal of Engineering and Technology*, 8(5), 2021.
- [4] Yihong He. Machine learning methods for credit card fraud detection. In *Highlights in Science, Engineering and Technology*, volume 23, Davis, CA, USA, 2022. Presented at IPIIS 2022.
- [5] Aditya Oza. Fraud detection using machine learning. Course project report, Stanford University, 2019.
- [6] Tzu-Hsuan Lin and Jehn-Ruey Jiang. Credit card fraud detection with autoencoder and probabilistic random forest. *Mathematics*, 9(21):2683, 2021.
- [7] Abdulwahab Ali Almazroi and Nasir Ayub. Online payment fraud detection model using machine learning techniques. *IEEE Access*, 2023.